

Mandi Operations & Staff Workflow Engineer

Focus: Mandi Staff Operator Command Center, Weighbridge & Quality Lab Entry

Executive Summary & Core Responsibility:

Developed the operational dashboard, Kanban stage queue lanes, weighbridge gross/tare logging, and DBT approval triggers.

1. Technology Stack & Architectural Components

Domain / Layer	Specific Technologies, Libraries & Frameworks Used
Dashboard Architecture	JavaScript Event Dispatchers, Real-Time Kanban Pipeline, Dynamic Modals
Mandi Operations UI	Tailwind CSS Cards, Interactive Status Pills, Counter Badges
Data Entry Modals	Dynamic Context-Aware Forms (Gate, Scale, Moisture Lab, Tare Re-weigh)
Live State Polling	Hybrid WebSocket sync with 1.5s background polling fallback
Audio & Voice Triggers	Automated bilingual voice approval announcements and stage chimes

2. Key Modules Developed & Production Deliverables

- **Mandi Operator Command Board (staff_dashboard.html & staff.js):** Built central operator interface with live counts for all 5 stages and mandi center switcher.
- **Kanban Pipeline Columns:** Developed drag/one-click transition lanes for Booked Slots, Gate Entry, Weighbridge, Quality Testing, and Completed DBT Payments.
- **Stage Data Capture Modals:** Built modal dialogs for digital scale weight inputs (Gross & Tare), moisture meter readings (FAQ Grade A standard $\leq 12\%$), and payment approvals.
- **DBT Payment Trigger:** Automated instant net weight computation ($\text{Net} = (\text{Gross} - \text{Tare})/100$) and unique DBT transaction ID assignment upon final tare entry.

3. Direct Codebase Ownership & Modified Files

Source Files:

frontend/templates/staff_dashboard.html • frontend/static/js/staff.js • backend/app/routers/staff.py

4. SIH Jury Defense & Live Evaluation Points (What to Speak Tomorrow)

- **Presentation Pitch Point:** Explain how the staff board eliminates manual paper registers and duplicate token chaos.
- **Presentation Pitch Point:** Demonstrate the one-click stage advance with Gross weight logging and instant farmer screen update.
- **Presentation Pitch Point:** Highlight the tamper-evident calculation where net weight and payout are calculated by server logic.